Public datasets



SIDTD (on MIDV-2020):
10 designs, text alterations



IDNet: Arizona, West Virginia;
face and text attacks



FantasyID: 13 designs, 10 scripts,
printed cards; test only

Data preparation

Unified records: design, identity, label, attack, fields, provenance

Rectify every page to its design canvas (512 x 800)

Design median reference + zone mask per design

Box masks from the altered fields; JPEG and size audit

Pairs (page, reference, zone); provenance-grouped splits

Proposed architecture

Base: ChangerEx (Open-CD), ResNet-18, LEVIR-CD initialization

Standard: difference fusion |f(page) - f(ref)| + f(page) + zone

Improved: attention fusion self-attn, cross-attn (Q = page, K,V = ref)

Heads: alteration map M and referral score s

Controlled baseline: the same network on the single page + zones

Training and evaluation

Loss: BCE(s) + 3 (BCE + Dice)(M) AdamW, three seeds

Augment: misalignment, independent JPEG on both inputs

Protocol A: SIDTD hold-out
Protocol B: FantasyID, zero-shot
Protocol C: IDNet WV <-> AZ

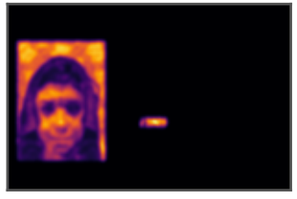
Metrics: AUC, EER, APCER@BPCER, pixel F1, transported threshold

CPU time per document, with cached reference features

Output



presented page +
design reference



deviation map M

refer / clear at a miss budget

released: references, masks,
splits, protocol, code